Transcriptomic Data Analysis on Diabetes Dataset

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1 Introduction

The purpose of this repository is to perform a transcriptomic data analysis on the dataset from Gene Expression Fulminant Type 1 Diabetes vs Classical Type 1A Diabetes from the Gene Expression Omnibus (GEO). The dataset contains an expression matrix of 10 samples and 41,000 genes and metadata about the samples. The goal of this analysis is to find any patterns or clusters and identify any features that may influence the observations.

2 Data Preprocessing

2.1 Metadata

We begin this analysis by preprocessing the metadata, containing information about each participant of the trial. The raw metadata contains a total of 40 columns, where some of these attributes appear multiple times. For this analysis, we only keep information about the diabetes type and age. The column are renamed and the age data is reformatted.

```
# Select columns from metadata that will be used for further analysis
metadata_preprocess <- as.data.frame(metadata) %>%
  rownames_to_column("sample") %>%
  rename(age = `age:ch1`, diabetes_type = `diabetes type:ch1`) %>%
  mutate(age = as.integer(gsub("yrs", "", age))) %>%
  select(sample, age, diabetes_type)
```

2.2 Gene Expression Matrix

The *data preprocessing* column of the raw metadata states *"Intensity values for each scanned feature were quantified using Agilent Feature Extraction Software, which performs background subtractions. Normalization was performed using Agilent GeneSpring GX version 11.0.2. (per chip normalization: 75 percentile shift; per gene normalization: median of all samples)"*. This tells us that the data in the given gene expression matrix is already normalised. We can verify that preprocessing has already been performed by inspecting the summary statistics of the gene expression matrix.

```
summary(expr_matrix)
```

```
#>    GSM1082954        GSM1082955        GSM1082956        GSM1082957
#>  Min.   :-4.9300   Min.   :-3.8900   Min.   :-4.1000   Min.   :-2.8900
#>  1st Qu.:-0.2800   1st Qu.:-0.2200   1st Qu.:-0.0900   1st Qu.:-0.1200
#>  Median : 0.0300   Median : 0.0400   Median : 0.0200   Median : 0.0300
#>  Mean   : 0.0608   Mean   : 0.0908   Mean   : 0.1005   Mean   : 0.0842
```

```
#>  3rd Qu.: 0.3600   3rd Qu.: 0.3400   3rd Qu.: 0.1900   3rd Qu.: 0.2200
#>  Max.   :10.4900   Max.   : 6.0200   Max.   : 6.9200   Max.   : 8.2200
#>  NA's   :14408     NA's   :13524     NA's   :15559     NA's   :14872
#>    GSM1082958        GSM1082959        GSM1082960        GSM1082961
#>  Min.   :-2.5300   Min.   :-3.9400   Min.   :-3.5100   Min.   :-2.8600
#>  1st Qu.:-0.0700   1st Qu.:-0.1800   1st Qu.:-0.3000   1st Qu.:-0.1600
#>  Median : 0.0400   Median :-0.0200   Median :-0.0400   Median :-0.0200
#>  Mean   : 0.1042   Mean   : 0.1414   Mean   : 0.1143   Mean   : 0.0914
#>  3rd Qu.: 0.2000   3rd Qu.: 0.2700   3rd Qu.: 0.3200   3rd Qu.: 0.1000
#>  Max.   : 6.2200   Max.   : 9.1700   Max.   : 6.7700   Max.   :10.4500
#>  NA's   :15059     NA's   :15476     NA's   :18966     NA's   :17242
#>    GSM1082962        GSM1082963
#>  Min.   :-3.8800   Min.   :-4.0100
#>  1st Qu.:-0.1300   1st Qu.:-0.1300
#>  Median : 0.0200   Median : 0.0400
#>  Mean   : 0.1014   Mean   : 0.0739
#>  3rd Qu.: 0.2300   3rd Qu.: 0.2400
#>  Max.   : 7.1500   Max.   : 6.4200
#>  NA's   :15100     NA's   :13639
```

Given the large number of genes, it is common to remove genes that have low expression values across samples. For each gene, we sum the values across all samples and only keep the gene if the resulting sum is above a threshold of 0.5. After filtering the genes, the matrix is reduced from 41,000 samples to 14,101 genes.

```r
mask <- rowSums(expr_matrix, na.rm = TRUE) > 0.5
expr_matrix_reduced <- expr_matrix[mask,]
```

The next preprocessing step is to handle missing values. The below table summarises the frequency of missing values for each gene.

```r
table(rowSums(is.na(expr_matrix)))
```

```
#>
#>     0     1     2     3     4     5     6     7     8     9    10
#> 20998  1785  1150   723   603   524   593   653  1058  3164  9749
```

The table shows that 20,998 genes have no missing values across the samples. For this analysis, we will only focus on the gene that have no missing values.

```r
expr_matrix_reduced <- as.data.frame(expr_matrix_reduced) %>% drop_na()
```

Finally, we will generate another dataframe by joining the expression matrix with the metadata by sample. Before we perform the join operation, we need to pivot the expression matrix into long form such that the samples are another attribute in the matrix.

```r
# Pivot expression matrix to longer form
expr_matrix_long <- as.data.frame(expr_matrix_reduced) %>%
  rownames_to_column("gene") %>%
  pivot_longer(cols = -gene,
               names_to = "sample",
               values_to = "normalised_count")
```

```r
# Combine expression matrix (in long format) with metadata
combined_data <- expr_matrix_long %>%
  left_join(metadata_preprocess, by="sample")
```

3 Exploratory Data Analysis

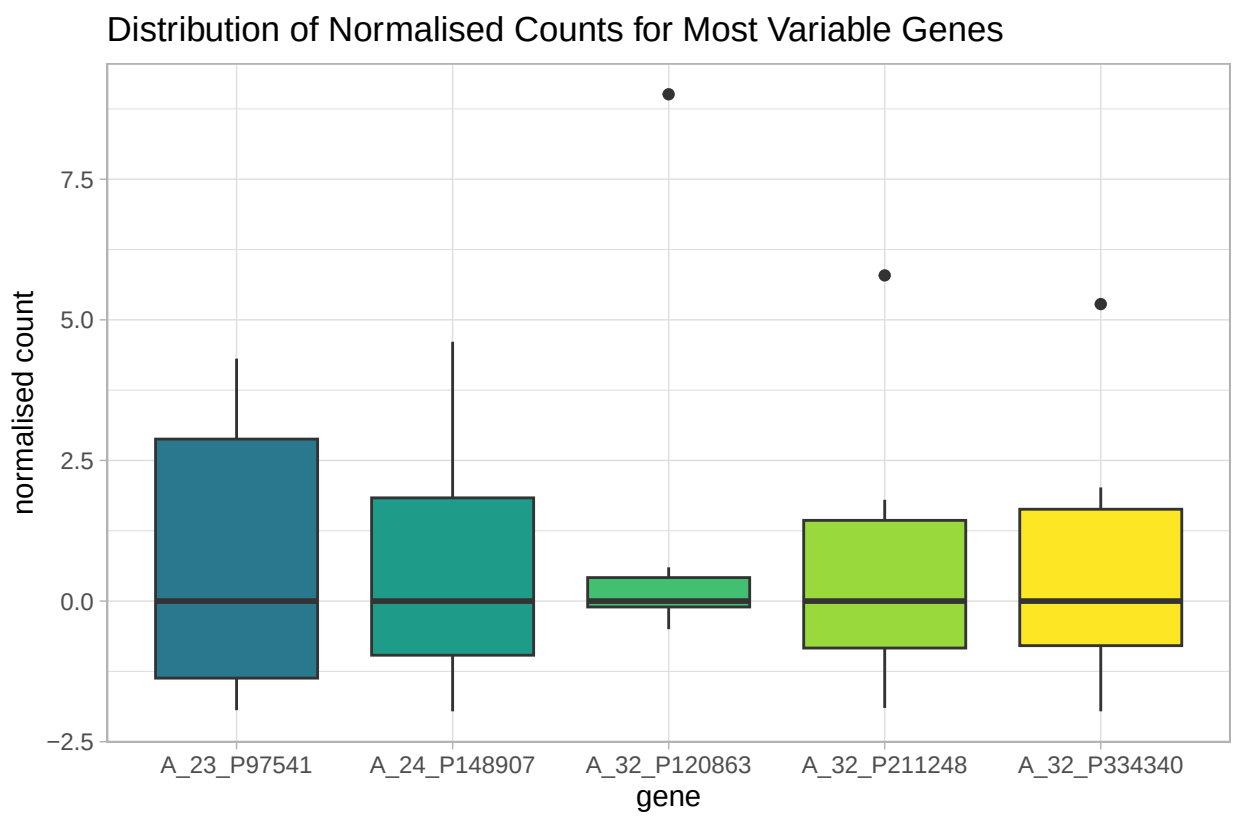
Using the preprocessed dataframes, we will conduct an exploratory data analysis to gain initial insights on the data.

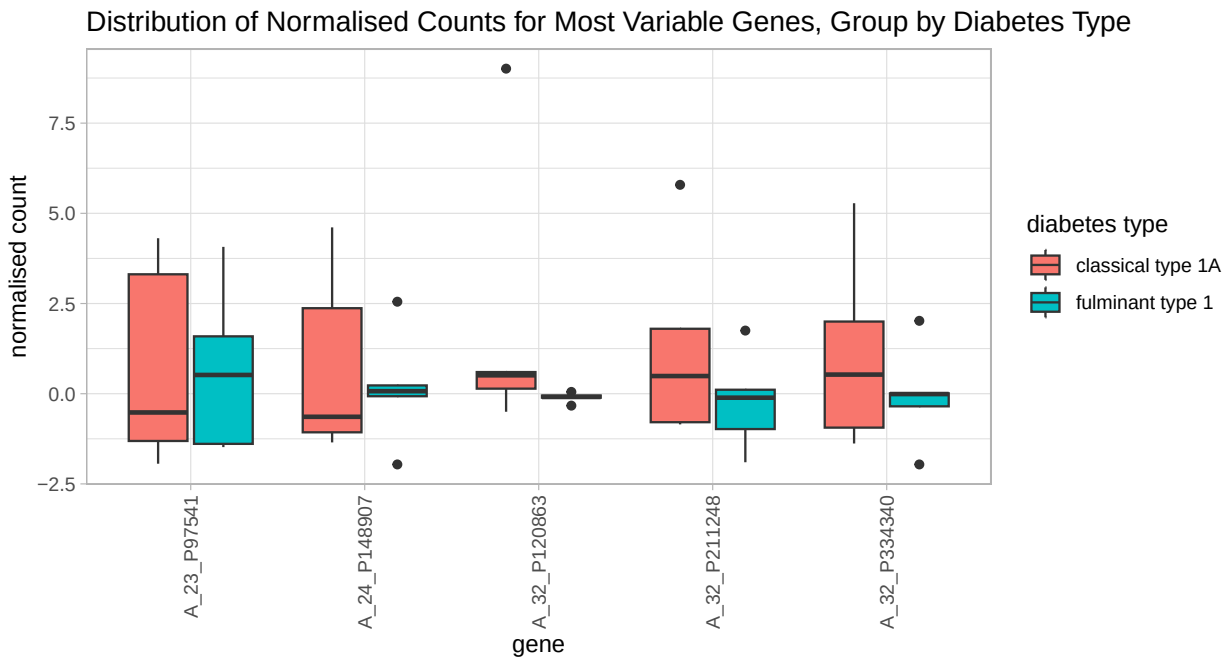
3.1 Normalised Counts against Most Variable Genes

The top 5 genes that have the largest variance are selected using the below R code.

```r
# Identify top 5 genes in dataset with the most variance
top_genes <- expr_matrix_reduced %>% apply(1, sd) %>%
  sort(decreasing = TRUE) %>%
  head(5)
```

We examine the distributions of normalised counts for the top 5 most variable genes. The below box-plots show the distribution of normalised count for each gene, where the second plot separates the distribution by the two types of diabetes in the dataset.

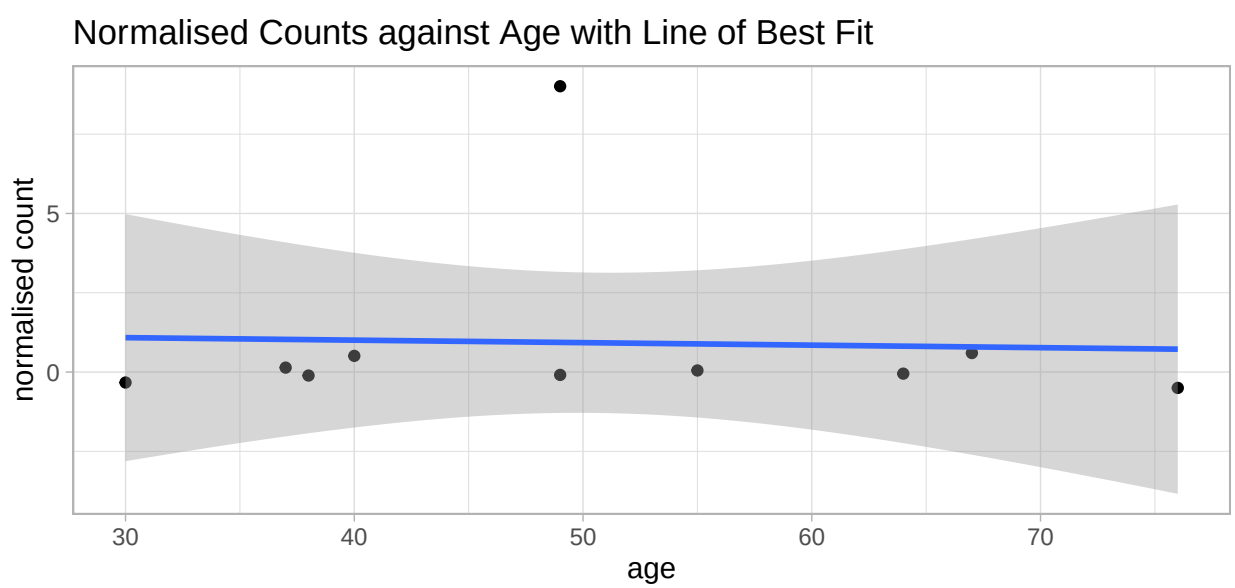


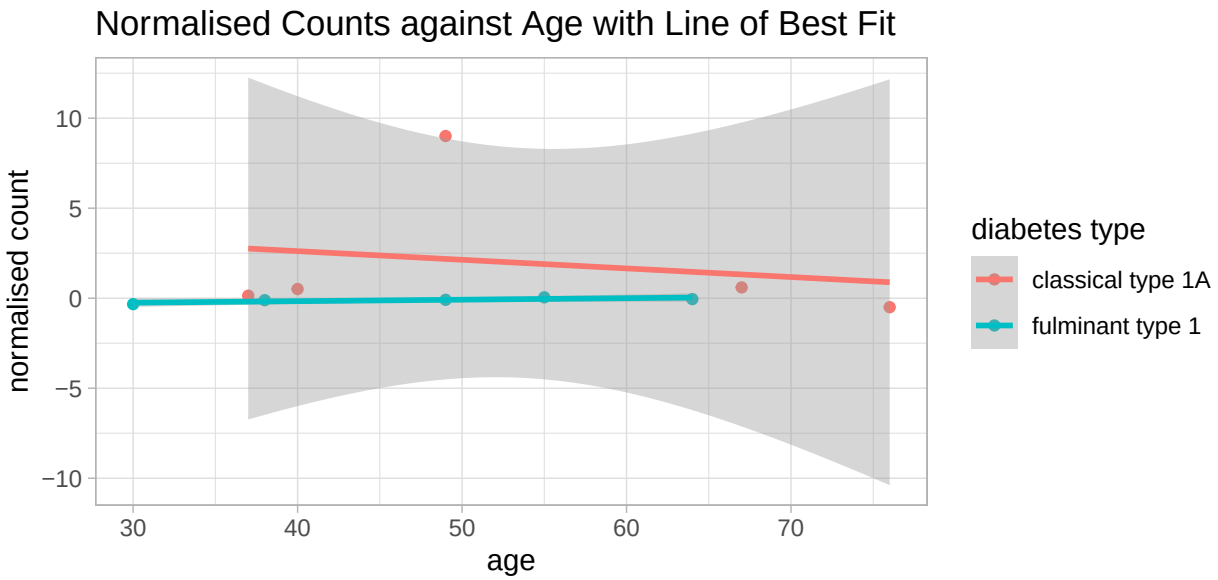


The box-plots shows that the distribution of normalised counts have some similar characteristics (ie: median, interquartile range) among the genes considered. An exception to this observation would be the distribution for gene *A_32_P120863* which appears to have a smallest interquantile range, but possesses a large outlier that influences the mean and variance of this distribution significantly. Separating the box-plots by diabetes type shows that the classical type 1A group have a larger variance compared to the fulminant type 1 group.

3.2 Normalised Counts against Age

We plot the the normalised count against the age of the participant. We restrict the data to points that are from the most variable gene in the preprocessed dataset. A line of best fitted with a 95% confidence band is included to visualise any trends and quantify uncertainty in the estimate line.

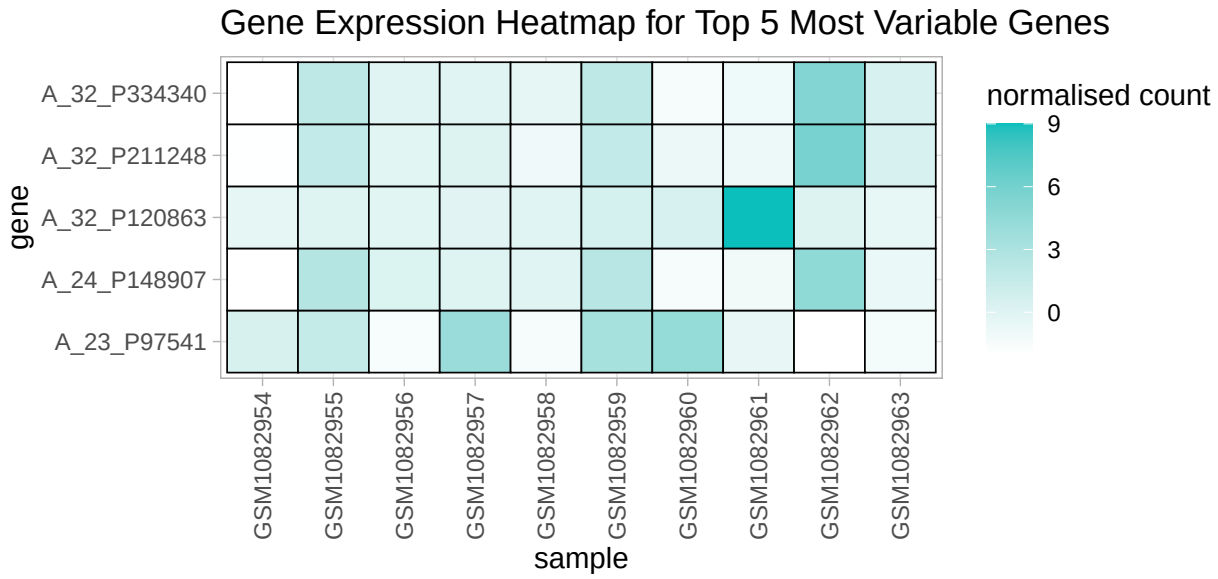




The normalised counts are fairly consistent across the different ages, with the exception of one sample who recorded a substantially large value with an age value of around 49. When the diabetes types are distinguished, we can see that the fulminant type group has a near-zero slope with a small confidence band, whereas the classical type 1A group has a negative slope with a large confidence band, which may be due to the outlier point distorting the line of best fit.

3.3 Normalised Counts across Samples

We conclude the exploratory analysis by generating a heat-map of the normalised counts for each sample, using the top 5 most variable genes. The 5 samples to the left of the plot are the fulminant participants and the 5 samples to the right are the classical type 1A participants.



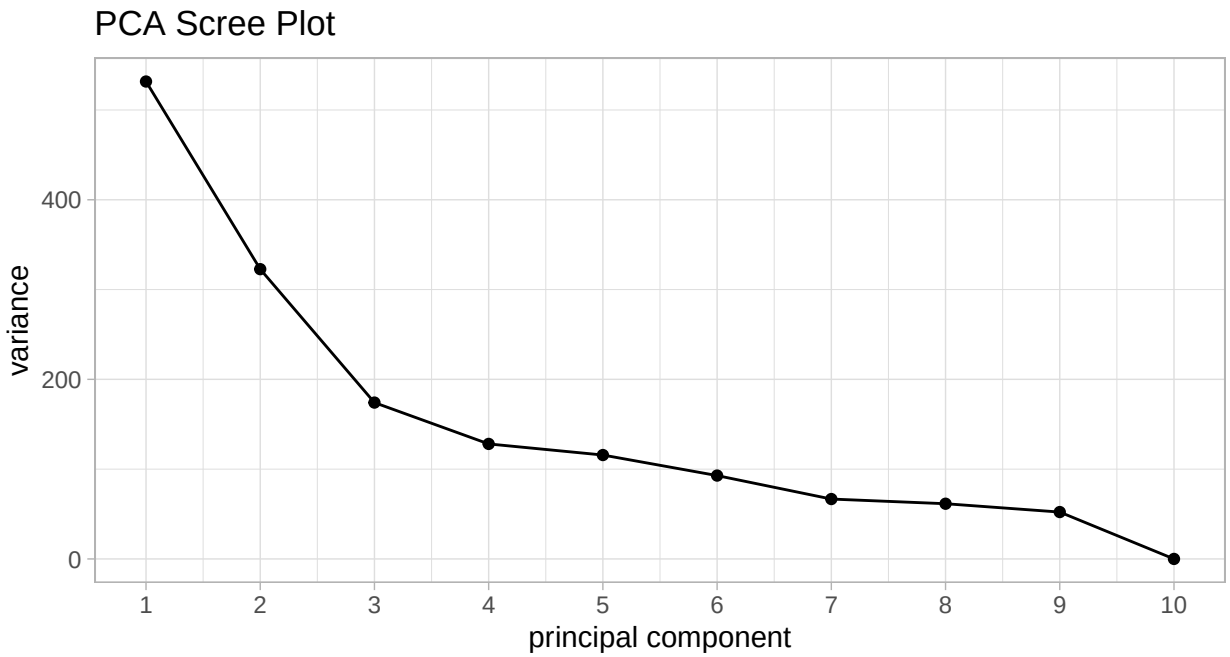
The fulminant participants appear to have a similar gene expression profile in this heat-map compared to the other group of participants. Sample *GSM1082961* seems to have a substantially different gene expression profile for the top 5 most variable genes compared to all other samples in this study.

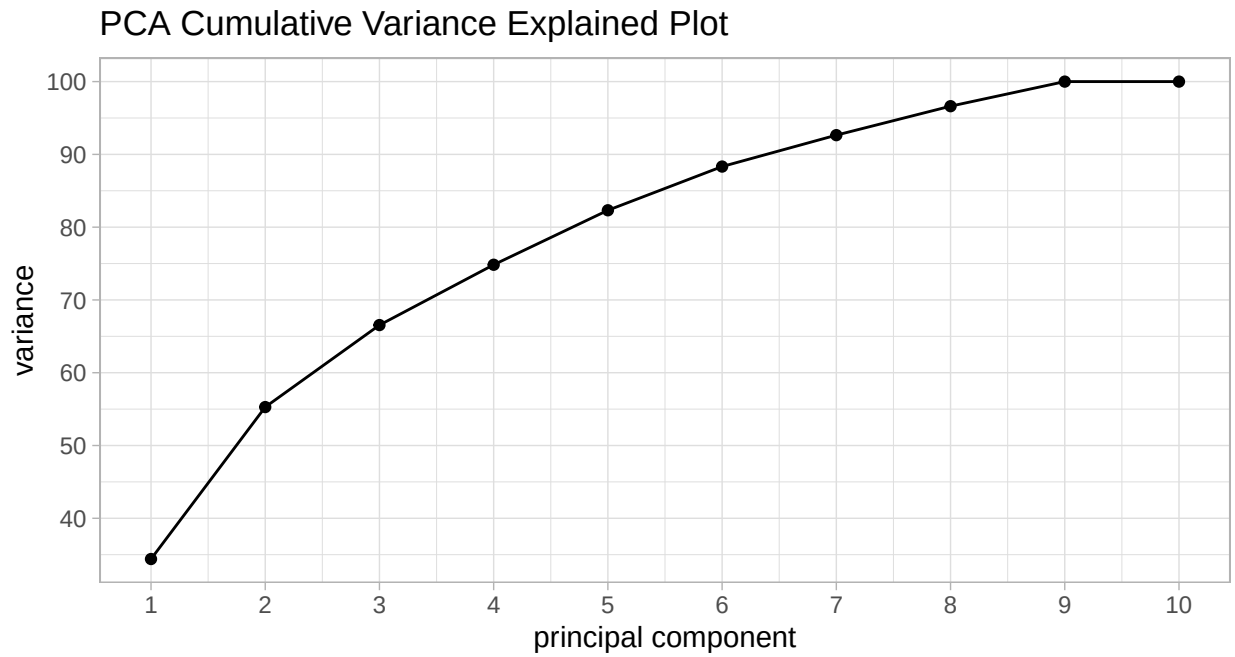
4 Dimensionality Reduction

In this section, we explore two dimensionality reduction techniques to generate a different representation of the gene expression data with much fewer dimensions. Dimensionality reduction is a common procedure when analysing genomic data to help identify patterns and features that influence certain genes.

4.1 Principal Component Analysis (PCA)

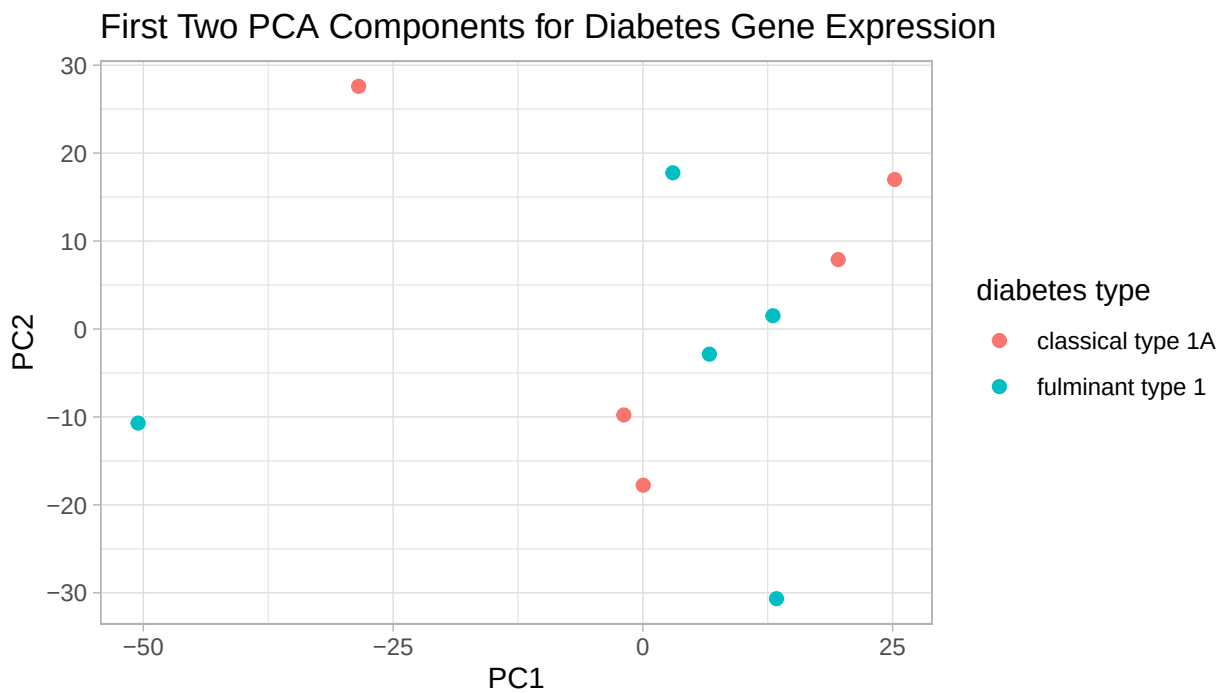
One of the most prominent approaches to dimensionality reduction is PCA. This transformation is a linear dimensionality reduction technique that aims to find new features that encompasses the most variance from the given data. To inspect how much variance how been captured by each component, we can use scree plots to visualise the variance of each component as a line graph. We also generate a plot of the cumulative variance explained against the number of PCA components to see how much of the variance has been captured over the number of components.





From both the scree plot and variance explained plot, the benefits of adding more component appears to diminish at the fifth component. Using PCA on this dataset, roughly 82% of the variance can be explained with five principal components, reducing the number of original gene features substantially.

We plot the first two components of the data on a scatter-plot and colour each point with its associated type 1 diabetes group.



The first two PCA components of this dataset does not reveal any obvious clusters. In particular, the diabetes type do not appear to form distinct clusters in this representation.

4.2 t-distributed Stochastic Neighbour Embedding (t-SNE)

One of the major drawbacks of PCA is that it is a linear dimensionality reduction technique. For high-dimensional data with more complex structure, it may be more appropriate to utilise a non-linear technique. One of these methods is t-SNE. This probabilistic method generates new components where similar points are close by and dissimilar points are further away in the lower dimensional space with high probability. We will plot the first two t-SNE components of the data on a scatter-plot and colour each point with its associated type 1 diabetes group.

While the scatter-plot of the points using the t-SNE components appears to be different to the PCA components, there does not appear to be cluster that separates the two diabetes group in the study.